Sahaj Bhadja

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Summary

Backend engineer with **2+ years of experience** building scalable microservices, REST APIs, and data-backed platforms with strong focus on software quality, operational reliability, and repeatable delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Designed and optimized multi-tenant relational schemas across **Azure SQL, PostgreSQL, and SQL Server** for **500+ users** and **10K+ structured records**, reducing dashboard latency by **71%** (**1.2s to 350ms**).
- Built distributed backend workflows with **FastAPI**, async job execution, and cloud storage integrations, processing **5K+ monthly jobs** with **99%+** reliability and **3x** better throughput.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Engineered production backend services for a self-serve grading platform handling **10K+ requests/day** using **Flask, MySQL, AWS RDS, and REST APIs**, with concurrent batch execution for **1K+** files.
- Implemented structured grading, rubric, and model-usage workflows for **150+ instructors**, improving turnaround time by **60%** while maintaining stable API behavior under peak load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered microservice-based software with reusable OOP modules and **20+** automated unit/integration test suites using **Jenkins, PyTest, and Mockito**, validating releases across Windows, Linux, and macOS.
- Optimized APIs and ETL pipelines with **AWS Glue, S3, and design patterns**, improving response times by **30%**, pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built a **microservice-based resource platform** with **MERN**, WebSockets, and **Azure SQL**, synchronizing **2M+ records** across locations and supporting **8** business workflows for allocation, billing, and notifications.
- Automated testing and delivery with **Jenkins, Docker, Jest, Mocha, Selenium, and Kibana-based monitoring**, increasing release validation to **95%+** while improving delivery reliability and observability.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built low-latency **Flask** booking and payment APIs over Firebase for a cross-platform product handling **50K+ records**, improving confirmation speed by **40%** and reducing manual refund handling by **70%**.
- Designed pricing, coupon, webhook, and analytics abstractions with **Databricks, Spark, and Spring Boot-based services**, sustaining sub-**200ms** booking flows and hourly operational reporting.

Skills

Languages: Java, Python, JavaScript/TypeScript, SQL, C++
Backend & Services: REST APIs, Microservices, Spring Boot, FastAPI, Flask, Node.js, Express.js, WebSockets
Databases: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, Firebase
Testing & DevOps: Mockito, PyTest, Jest, Mocha, Selenium, Jenkins, Docker, CI/CD, Kibana, Monitoring
Engineering: Distributed Systems, Async Job Processing, ETL Pipelines, Debugging, Code Reviews, Agile

Projects

Chat2SQL GitHub | *Java, TypeScript, Docker, PostgreSQL*

- Built a SQL development workspace supporting **10+** databases, enabling schema exploration, structured query execution, and cross-service development across a **Java** and TypeScript codebase.

FinNexus Live | *Next.js, Kubernetes, MongoDB, WebSockets*

- Engineered a high-volume platform for **50K+ users** and **1M+ transactions**, building secure transaction workflows, real-time synchronization, and CI/CD-driven delivery for event-heavy application behavior.

Big-Brain Live | *PostgreSQL, FAISS, Pinecone, REST APIs*

- Built a document retrieval platform over **50K+ records** with sub-second search, metadata persistence, and incremental indexing, reducing retrieval latency by **40%** for scalable user-facing search workflows.